

Team and Contribution Record

Member	Area Owned	What That Involved
Roman Ali	Dataset generation, data quality and cleaning	Created the data simulation, provided each route with individual behaviors, included deliberate errors within the data, and wrote the verification and correction procedures to correct them.
Hamza Mughal	Spark platform, ingestion, integration and storage	Configuration of Spark (solving issues with Windows), loading data strictly with the schema, writing of the Spark SQL join query, and connecting with the storage (Parquet and HDFS) were done.
Warisha Rehman	Feature engineering and analytics	Designed the features that were utilized throughout the project, calculated and ranked the routes, identified the problematic stops, and analyzed passenger movement and bus bunching issues.
Bisma Batool	Machine learning, forecasting and model governance	Two models for overcrowding were created and compared, delays were predicted along with the level of delay severity, the accuracy of the prediction was determined, demand and occupancy were forecasted, routes were clustered and model versions were controlled
Abdul Moeed	Recommendations, serving, access control and documentation	Recommendation and what-if scenario generation rules were created, both interfaces were developed, login/permission controls were developed, tests were written and a project report was generated